

EE / CprE / SE 491 — sdmay27-11

Weekly Report 3:

MIMO FMCW Radar

28th September – 3rd October 2026

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§1 Summary

This week the team focused on reducing risk and scoping the system down to a minimum viable product. The original DACs were overkill for the application, so we designed a new, simplified DAC for sweeping the VCO. On the bench, we tested the ADAR2001 and ADAR2004 beamformers, wrote Python scripts to program them for optimal harmonic rejection, and swept the lab VCOs using an NI DAQ USB device. That testing exposed a negative-voltage fault that forward-biased the lab VCO's varactor diode, so negative-voltage protection will be added to all of our circuits. The VCO board was revised to a single output so that a PLL/synthesizer can follow it, and the TX up-converter output traces are being simulated in CST to set their characteristic impedance. The Vivaldi antenna is still being evaluated as the microstrip-to-waveguide transition but is not yet matched across 24 to 40 GHz. Signal processing can now pull samples from the lab spectrum analyzer into Python and compute FFTs, and the FPGA sub-team set up an Opal Kelly board with a blinky test, planned the SPI controller, and weighed an MCU-based alternative.

§1.1 Team and Roles

Team Member	Role
Flatrim Dreshaj	Team Leader
Justin Adrowski	FPGA Programming
Durotimi Johnson	TX/RX Layout Design
Ryland Feist	VCO / PLL / Synthesizer
Elliott Bourgeois	Signal Processing
Brett Brunner	Antenna Design
Aidan O'Grady	FPGA / MCU Programming

Table 1. Team members and roles

§1.2 Status at a Glance

Work Item	Owner	Status
ADAR / VCO / DAC bench test	Flatrim Dreshaj	In Progress
VCO board	Ryland Feist	Revised to single output; PLL integration under study
TX up-converter board	Durotimi Johnson	Schematic complete; layout in progress (CST trace simulations)
Waveguide and transition design	Brett Brunner	In progress
Antenna design	Brett Brunner	In progress
IF capture chain	Elliott Bourgeois	Spectrum analyzer capture and FFT working in Python; DSP algorithms next
FPGA SPI controller	Justin Adrowski, Aidan O'Grady	Opal Kelly board up; SPI controller planned; MCU alternative under evaluation

Table 2. Status of each work item at the end of the week

§3 PCB Design

This week, the PCB design team worked on VCO temperature dependence and continued work on the ADAR TX. Separately, we did a lot of bench testing of the ADAR2001 and ADAR2004, wrote Python scripts to program the ADARs for optimal harmonic rejection, tested the VCOs we have in the lab, and programmed an NI DAQ USB device to sweep the VCO. During this testing we found an issue with the lab VCO: a negative voltage had accidentally turned on the varactor diode, which clamped the VCO tuning voltage to the diode's forward-bias voltage. Next week we will add negative-voltage protection to all of our circuits.

§3.3 VCO Board

This week the VCO board was revised so that the output no longer splits and instead remains a single output. We now plan to feed the VCO into a PLL/synthesizer, so the split will happen after the PLL rather than at the VCO output. We also continued looking into how a synthesizer would be integrated, and considered adding a heater to the VCO chip so that it always operates at a relatively consistent temperature.

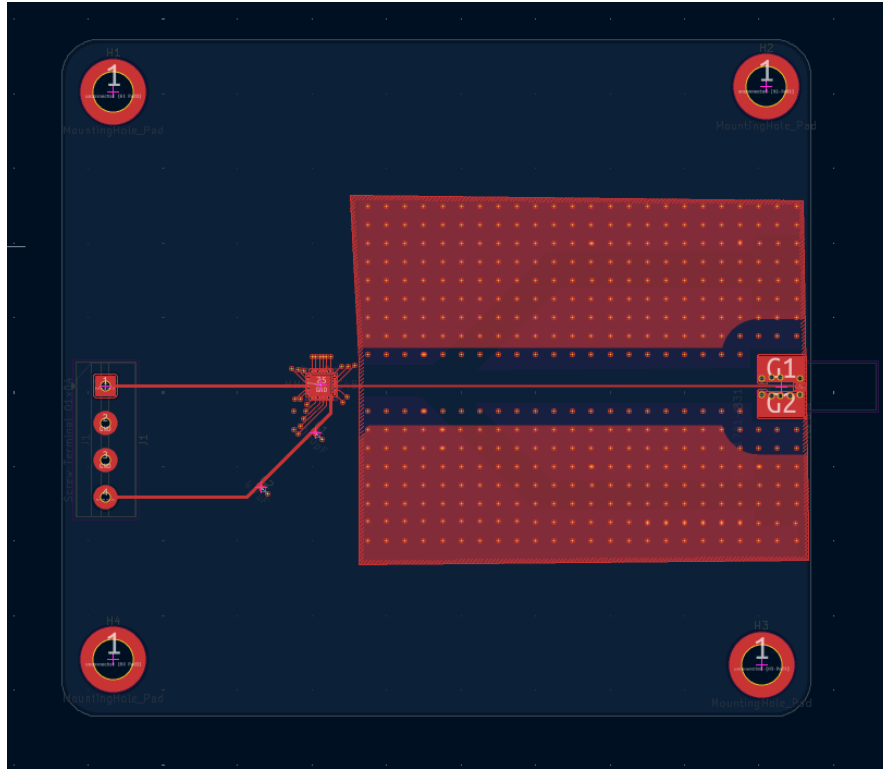


Figure 1. Revised VCO board layout

§3.4 TX Up-Converter Board

The TX board layout has not changed much this week because the output routing needs to be handled carefully. We are using CST Microwave Studio to simulate the coupling between these traces and choose a characteristic impedance based on the results.

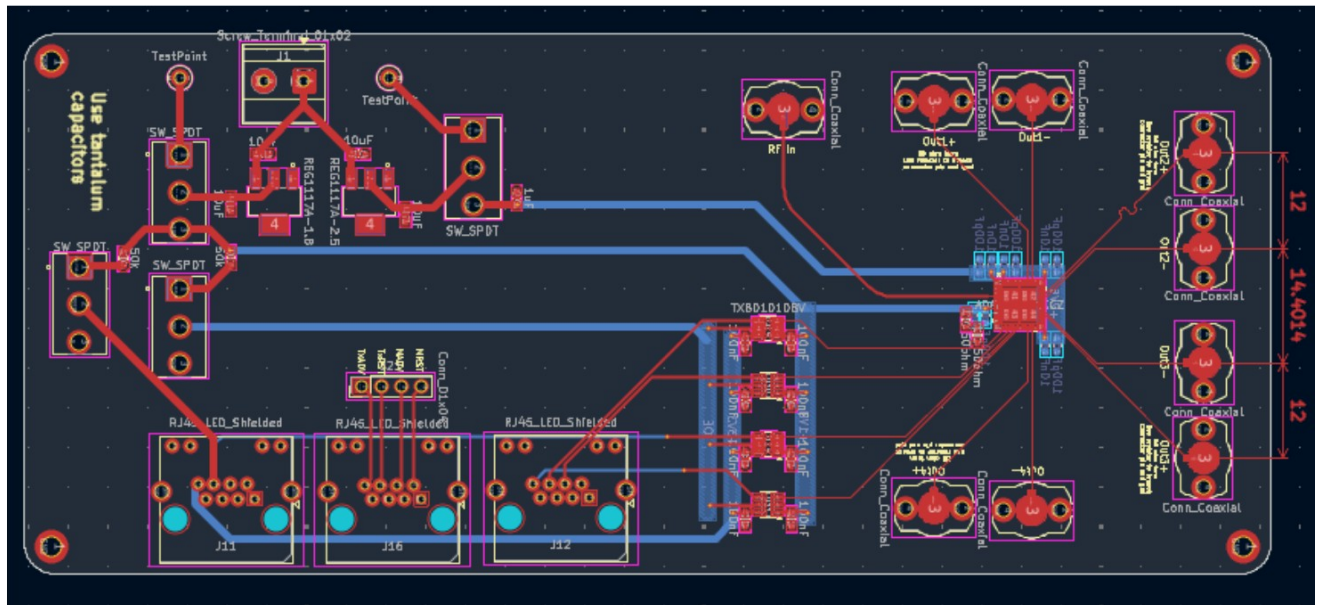


Figure 2. TX up-converter layout with temporary output traces

The negative outputs are all routed to a trace length of about 23.0 mm. The positive outputs will be terminated with a matched 50 Ω load, but since we may still use them, we plan to place a 50 Ω resistor from the output pad to the ground pad. That way the resistor can be removed and the positive output used another way. With that in mind, the positive-output traces were also matched to about 27.7 mm.

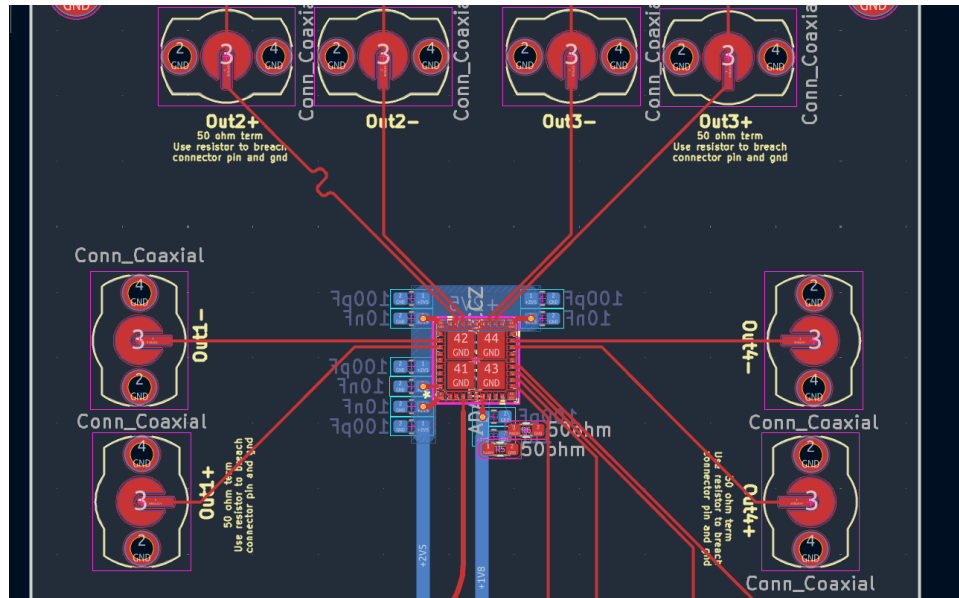


Figure 3. TX up-converter layout (close-up)

§4 Waveguide and Antenna Design

The Vivaldi antenna needs a microstrip-to-slotline transition, and several design variables have been adjusted to improve S11. Additional features (a stub and a transition) are needed to further improve the Vivaldi's performance, and the waveguide port setup in CST needs adjustment.

§4.1 Waveguides and Transitions

This week covered research into waveguide cutoff and microstrip-to-waveguide transitions (including tapered-slot and finline types), along with antenna properties, to guide the design in CST.

§4.2 Vivaldi as a Waveguide Transition

The goal continues to be S11 below -10 dB from 24 to 40 GHz; the best result so far is about -6.4 dB, with the worst around -2 dB. A stub and a transition still need to be added, along with other parameter tweaks. The optimizer will be used once the antenna's core

design has been modified. A quick prototype patch antenna will also be designed as an alternative candidate.

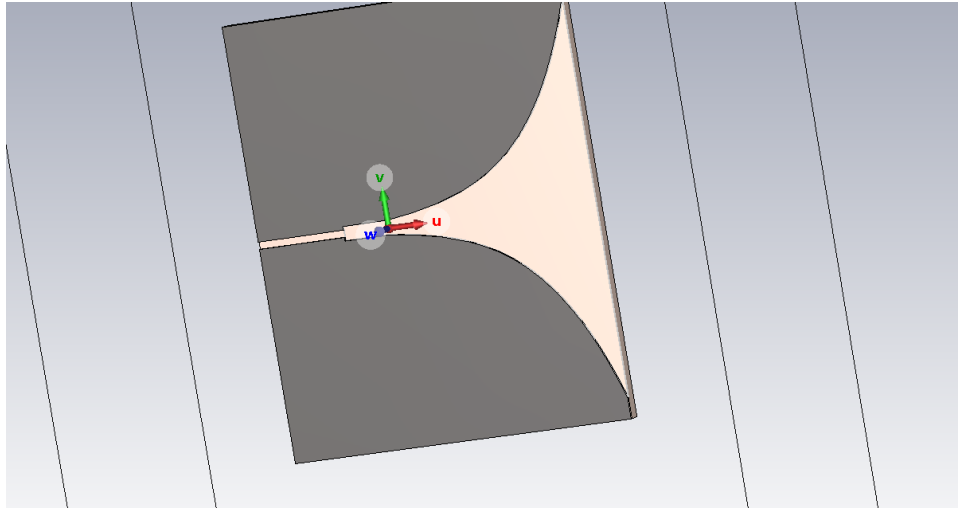


Figure 4. Vivaldi #1: wider flare, but some parameter sizes are wrong and need adjustment.

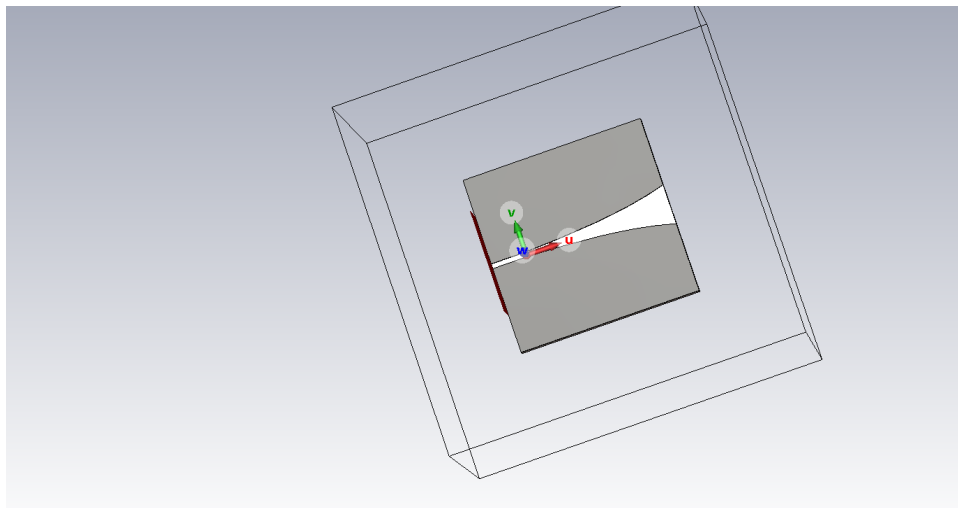


Figure 5. Vivaldi #2

§5 Signal Processing

Until the FPGA ADC path is running, the lab spectrum analyzer will sample the IF signal from the receiver board and pass it to a PC (Figure 6). This lets algorithm development start now.

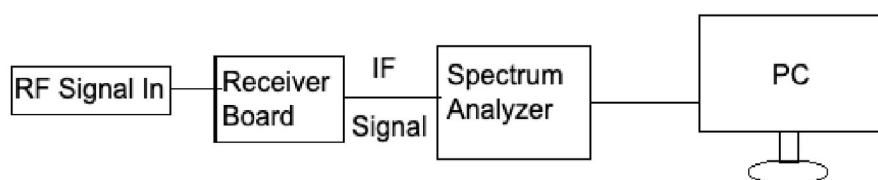


Figure 6. Planned IF capture chain using the lab spectrum analyzer

This week we set up the interface between the spectrum analyzer and the PC through its API. The API is written for C, so we use Python's ctypes with a .py wrapper to drive the spectrum analyzer from Python. So far we can collect samples from the spectrum analyzer and compute FFTs on them.

This process involves communicating with the spectrum analyzer, setting the center frequency and bandwidth, as well as the sample rate/communication rate, then sampling for a fixed number of samples. All of these are configurable via Python, allowing automation of data collection to test DSP algorithms before the rest of the system is functioning.

Next, we will explore our backup method of data acquisition, an NI DAQ card, which acquires data through its integrated ADCs. This will serve as an alternative to the microcontroller approach we are also exploring.

§6 FPGA Development

The FPGA sub-team set up an Opal Kelly board with a blinky test and planned the SPI controller. The plan is to use SPI to interface with an STM32 configured as an SPI slave and print the results to PuTTY or another serial console.

There is a fundamental trade-off between an MCU-based approach and an FPGA-based one. The FPGA has a much higher setup cost but much more predictable timing. Setting up and demoing an STM32 with SPI is almost out of the box, but more work is needed to meet timing, since the ADC channels must be sampled on the same sample instant at 1 MS/s. It is much harder to guarantee that serialized reads land on the same sample, or that DMA is set up to capture in parallel. Because the MCU board is also much easier to set up, we will try that first and see what issues arise.

§7 Research

- **SPI controller architectures:** reviewed for the VHDL design.
- **PLL research:** for the VCO and synthesizer.
- **Waveguides and transitions:** waveguide cutoff and microstrip-to-waveguide transitions, including tapered-slot and finline types.

- **Vivaldi design:** how the flare geometry sets the lowest usable frequency.

Bench measurements this week were limited to ADAR and VCO testing (§3) and the first spectrum analyzer captures (§5). Full system data collection starts once the first boards are fabricated and assembled.

§8 Pending Issues

- **Board fab and assembly:** first boards still need to be fabricated and assembled.
- **Waveguide transition:** the Vivaldi is a candidate, not yet final, and must match well at 24 GHz.
- **Vivaldi match:** S11 is not yet below -10 dB across 24 to 40 GHz (best about -6.4 dB, worst about -2 dB).
- **ADAR placement:** still open. It sets the RF trace lengths for the eight equal-length TX outputs.
- **TX up-converter:** the schematic may change, and the layout is not finished.
- **Negative-voltage protection:** needs to be added to all circuits after the lab VCO varactor issue (§3).

§9 Individual Contributions

Team Member	Contribution	Weekly Hours	Total Hours
Flatrim Dreshaj	ADAR, DAC and VCO eval board system testing	10	60
Brett Brunner	Antenna design	5	21
Durotimi Johnson	TX up-converter board	4	30
Ryland Feist	VCO board, PLL research	3.5	13.5
Elliott Bourgeois	Signal processing	6	14
Justin Adrowski	FPGA	3	7
Aidan O'Grady	FPGA	4	12
Team Total		35.5	157.5

Table 3. Individual contributions and hours

§10 Plans for Next Week

- **PCB (Flatrim):** Run CST simulations of the differential lines, connectors, and terminations for the ADAR board, and add negative-voltage protection to our circuits.
- **Lab equipment (Flatrim):** Fix lab equipment for more bench testing.

- **Antenna (Brett):** Research antenna design techniques, create the microstrip-to-slotline transition, adjust the slot cavity, add the stub, and design a prototype patch antenna.
- **Signal processing (Elliott):** Start DSP algorithm development and explore the NI DAQ card.
- **TX up-converter (Durotimi):** Update the trace widths to get desired characteristic impedance based on coupling simulation results.
- **FPGA (Justin, Aidan):** Check the feasibility of an MCU-based approach for signal processing, and work out how to send data to the “home base” PC for FFT post-processing.
- **VCO (Ryland):** PLL board schematic and research.